

Azadirachta indica (Meliaceae)

Neem, Margosa Tree

Neem is one of the most valued herbs in Indian and Ayurvedic medicine. Extracts of the leaves are used to treat conditions such as asthma, eczema, diabetes, and rheumatism, while neem oil has been applied as a hair lotion, to treat headlice, and to calm angry skin rashes. Research indicates that neem may prove useful as an insecticide and a contraceptive. The tree itself is said to purify the air and is widely planted in India.



Neem has so many medicinal uses that the tree is considered a "natural pharmacy."

Habitat & Cultivation

Native to Iran, Pakistan, India, and Sri Lanka, neem is found throughout the subcontinent in forests and woods, often being planted on roads to provide shade. It is now naturalized in other tropical regions, including Malaysia, Indonesia, Australia, and West Africa. It is grown from seed. Leaves and seed are harvested throughout the year.

Related Species

Melia azedarach, also an Indian plant, is a very close relative with particular value in treating intestinal worms. It is often used as a substitute for neem.

Key Constituents

- Meliacins
- Liminoids
- Triterpenoid bitters
- Sterols
- Tannins
- Flavonoids

Key Actions

- Anti-inflammatory
- Lowers fever

- Antimicrobial
- Promotes wound healing
- Antiparasitic
- Antimalarial

Research

■ **Recent research** This indicates that neem oil is both anti-inflammatory and antibacterial, and to some degree reduces fever and lowers blood-sugar levels.

■ **Insecticide** Extensive research shows that liminoid azadirachtins are insecticidal and inhibit feeding and growth—making neem an inexpensive and ecologically sound insecticidal agent. The azadirachtins are also linked to the tree's antimalarial activity.

■ **Diabetes** Research indicates that neem leaf and oil act to stabilize blood-sugar levels and may be helpful in treating or delaying type 2 diabetes.

Traditional & Current Uses

■ **Medicine chest** Thought of in India as almost a pharmacy in its own right, all parts of the neem tree may be used medicinally. The bark is bitter and astringent and a decoction is used for hemorrhoids. The leaves are taken as an infusion for malaria, peptic ulcers, and intestinal worms, and may be applied locally as a juice, infusion, or ointment to skin problems including ulcers, wounds, boils, and eczema. The juice of the leaves is also applied to the eyes to treat night blindness and conjunctivitis. The twigs are used as a tooth cleanser, firming up the gums and preventing gum disease.

■ **Neem oil and sap** Neem oil, expressed from the seeds,

**Neem**

An evergreen tree, growing to 52 ft (16 m), with compound leaves and white flowers.

Parts Used

Leaves can be infused and applied as a lotion to many skin rashes, including acne, eczema, and psoriasis.

Twigs are used as toothpicks to improve oral hygiene.

Seeds contain neem oil, which can be expressed and used to treat headlice in adults and children.



Fresh leaves

Fresh seeds

Key Preparations & Their Uses

⚠ **Cautions** Do not take during pregnancy, while breastfeeding, or during fertility treatment. In children, use topically only. Long term high-dose treatment is not advisable.



Oil from the seeds is used to prevent and treat infestations of headlice. Apply daily to the scalp.



Cream (to make, p. 296) is used to treat many skin disorders, including eczema. Apply as required.



Tincture (to make, p. 292) is bitter and can be used to treat fevers.



Infusion of the leaves (to make, p. 291) can be used for conjunctivitis.



Decoction of the bark (to make, p. 291) is strongly bitter and astringent.

is commonly used as a hair dressing and is strongly antifungal and antiviral, preventing scabies and ringworm, among other things. It can be made into a useful and easily applied treatment for headlice. The oil is also used to treat skin

conditions such as eczema, psoriasis, and even leprosy, and as a vehicle for other active ingredients. Neem oil should be avoided when attempting to conceive a child as it can reduce fertility in both women and men.